VANSH MITTAL

mittalvansh21@gmail.com | +91 9557288779 | Noida, India | LinkedIn.com/in/vanshmittal18 | GitHub.com/vanshhh18

PROFESSIONAL SUMMARY

Final-year Computer Science (Data Science) undergraduate specializing in AI agent engineering. Building, orchestrating, and evaluating LLM-powered agents with production-grade guardrails across orchestration (LangGraph/LangChain), tooling (FastAPI, REST APIs), and evaluation (rubric design, edge-case testing). Proficient in Python, scikit-learn, and deep learning, with a strong foundation in probability, statistics, and linear algebra.

TECHNICAL SKILLS

Languages: Python, SQL, JavaScript
AI Agents & LLM: LangChain, LangGraph, RAG, FAISS, Hugging Face Embeddings, Prompt Engineering, Agent Orchestration & Tooling
ML / Data Science: Pandas, NumPy, scikit-learn, PyTorch, TF-IDF, NLP, Power BI
Engineering & Deployment: FastAPI, REST APIs, Docker, AWS, Git, GitHub, Postman, Vector Databases
Evaluation & Guardrails: Rubric design, edge-case & regression testing, human-in-the-loop approval workflows
Mathematics: Probability & Statistics, Linear Algebra, Calculus, Data Structures & Algorithms

INTERNSHIP & EXPERIENCE

AI Systems Evaluation Intern - AirDawg Labs *July 2026 – October 2026 | Remote*

- Authored 19 tasks for Terminal-Bench 2.1 and 3, an LLM agent benchmark (8-9 accepted after review), each with a verifiable test harness targeting reasoning gaps in CLI and Linux problem-solving.
- Designed correctness and efficiency rubrics, revised tasks from reviewer feedback, and stress-tested them against frontier models (GPT 5.6, Opus 5) to check difficulty and surface failure modes.

AI Intern - DEAL, DRDO *June 2026 - July 2026 | On-site*

- Built a production RAG pipeline for an SDR system, processing 10,000-100,000 defence-relevant records with intelligent retrieval and analysis.
- Engineered agentic workflows using LangGraph to orchestrate multi-step decision-making, improving retrieval accuracy and cutting manual intervention in signal data handling.

PROJECTS

RecoverAI - AI-Powered Payment & Revenue Recovery System GitHub *August 2026*
Python, Scikit-learn, Razorpay API, Groq LLM, Streamlit

- Built an end-to-end ML payment recovery pipeline using a Decision Tree classifier (88.9% held-out accuracy) and adaptive policy engine to diagnose six failure causes and dynamically select recovery actions.
- Integrated Razorpay APIs, guarded retries, payment links, and AI voice/chat recovery, achieving 61.0% recovery vs. 40.4% with naive retry and recovering ₹8.05L vs. ₹5.33L across 180 held-out payments.

OpsCopilot - LLM Agent with Human-in-the-Loop Approval GitHub *July 2026*
Python · FastAPI · SQLAlchemy · LangGraph · LangChain · Pydantic · SQLite · Streamlit · Groq

- Orchestrated a LangGraph ReAct agent (Qwen3-32b) with 12 tools spanning read and write operations, achieving 100% intent-to-tool accuracy across 12 test queries at ~1.5s average latency.
- Built an approval-gated guardrail workflow requiring explicit human confirmation before mutating actions, correctly blocking or safely handling 80% of invalid inputs (nonexistent users, negative values, approval bypass attempts) in edge-case testing.

Multi-Agent AI Helpdesk System with NLP Classification & RAG GitHub *June 2026*
Python · FastAPI · scikit-learn · LangGraph · ChromaDB · Sentence-Transformers · Groq · HuggingFace

- Trained a LinearSVC classifier on 25,000+ support tickets with TF-IDF vectorization, achieving 97% accuracy / 96% F1 across 10 departments and 3 priority levels; deployed via FastAPI on HuggingFace Spaces.
- Engineered a multi-agent RAG workflow (LangGraph + ChromaDB + Sentence-Transformers) for context retrieval and automated, domain-specific end-to-end ticket routing.

CERTIFICATIONS

Artificial Intelligence Fundamentals : IBM SkillsBuild, 2026 ; Agentic AI Certified Foundations Associate : Oracle, 2026

ACHIEVEMENTS

Selected as a Top 6 finalist among 100+ participants in an AI Automation competition hosted by UpNex Solutions.

- Designed and built an automated lead-generation agent using n8n during the final round and won a $50 cash prize.

EDUCATION

B.Tech : Computer Science & Engineering (Data Science) | CGPA: 7.32/10 (Semesters 1-6) | 2023-2027
ABES Engineering College, Ghaziabad, India
Higher Secondary (PCM) : 88% | 2022-2023